Security and Privacy for Machine Learning (Honors)

Internal Assessment 2

Implementation, Evaluation and Technical Outcomes

Backdooring the Guard

A Data-Poisoning Attack on a CAN-Bus Intrusion Detector,
and a Layered Privacy Defence (DP-SGD, Local DP, Compressive Privacy)

Name	Aishwarya Nitin Gawali
Roll No.	16014223007
Batch	B1
Assessment	IA-2 (20 marks)

A backdoor attack on a CAN intrusion detector, and course privacy mechanisms studied as both defences and measured utility costs.

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Rubric-compliance map. The report mirrors the five IA-2 criteria.

IA-2 criterion	Evidence	Section
Problem refinement	Changes-since-IA-1 table, objectives, research questions	§1
Integration of 5 modules	Architecture, final mapping, per-module implementation	§2
Dataset & implementation	OTIDS provenance, windowing, individual contribution	§3
Evaluation & analysis	Backdoor, DP, LDP, compressive, coverage, limitations	§4
Final documentation	Conclusions, reproducibility, references	§5

1 Problem Refinement and Objectives

An in-vehicle Controller Area Network (CAN) has *no message authentication* (Koscher et al., 2010), so machine-learning intrusion detectors (IDS) are deployed to catch injected frames. But the IDS is itself a model trained on logged traffic, and its training pipeline is a *supply-chain attack surface*: a poisoned trigger can make the deployed IDS wave real attacks through on command (Gu et al., 2017). IA-1 established the attack pipeline on the HCRL CAN data and a first backdoor demonstration. IA-2 completes the backdoor on a 1-D CNN, adds the differential-privacy defence with a full ε sweep, and extends the study with local DP, output modification and compressive privacy. Every refinement is driven by an IA-1 finding (Table 1).

Table 1: Changes from IA-1 to IA-2 and their motivation.

Change in IA-2	Motivating IA-1 finding	Evidence
Backdoor on the 1-D CNN (not only RF)	IA-1 CNN washed out a 4-frame trigger under global pooling	§4.2
Strengthen the trigger (k=12 frames) and sweep poison rate	IA-1 showed the backdoor was unstable at low poison	Fig. 2
DP-SGD with a full ε sweep + ASR-vs-ε	IA-1 established DP-SGD only conceptually	Fig. 3
Membership inference on the IDS, vs ε	IA-1 named the privacy threat but did not measure it	Table 5
Local DP on CAN features; LDP Naive Bayes	IA-1 treated collection-time privacy analytically	Fig. 4
Output modification + compressive PCA/DCA	IA-1 listed these as remaining module coverage	§4.5
Class-balanced training throughout	IA-1 attack windows dominated the captures	§4.2

Research questions. (RQ1) Can a trigger-based backdoor make a near-perfect CAN-IDS misclassify attacks as normal, while staying invisible on clean traffic? (RQ2) Does differentially private training reduce the backdoor's attack-success rate, and at what detection cost? (RQ3) How do local DP and compressive privacy trade IDS utility for protection of the raw telemetry?

2 Integration of the Five Modules

Every mechanism is anchored to the *same* CAN-IDS; nothing strays from the backdoor problem. Figure 1 shows the pipeline.

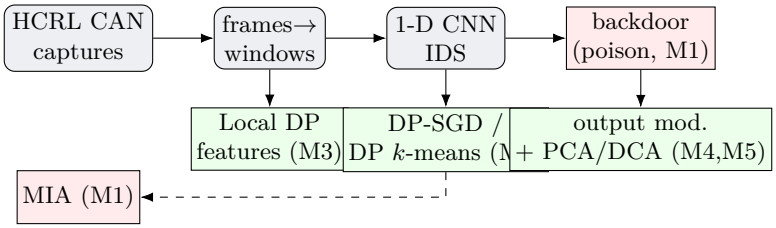


Figure 1: All five modules anchored to one CAN intrusion detector.

Table 2: Final module mapping.

Module	Role on the CAN-IDS	Evidence
M1	Backdoor/poisoning attack (headline threat); membership inference	§4.2, Table 5
M2	DP-SGD private training (ASR-vs-ε, MIA-vs-ε); DP k-means	§4.3
M3	Local DP perturbation of CAN features (Laplace/Duchi/Piecewise); LDP-NB	§4.4
M4	Output modification as an MIA defence; CAN-ID k-anonymity (analytical)	§4.5

Module	Role	Evidence
M5	Compressive PCA/DCA of the 1100-dim windows	§4.5

3 Dataset Finalisation and Individual Implementation

Dataset. The HCRL in-vehicle CAN intrusion captures (OTIDS family), via the public Kaggle mirror `bikashkundu/can-hcrl-otids`: one normal capture and three attack captures (`dataset.csv`, `dataset1-3.csv`; columns `TS,ID1,ID0,LEN,DLC0-7,target`). Full provenance – source, access method, per-file row counts and SHA-256 hashes, licence note – is in `data/PROVENANCE.md`.

Representation. Each frame becomes $[\,\Delta t,\ \text{CAN ID},\ \text{DLC},\ b_0 \ldots b_7\,]$ (11 features); a window of 100 consecutive frames (stride 50) is one example, labelled *Attack* if it comes from an attack capture, with a time-contiguous split *inside* each capture so no window crosses the train/test boundary. This yields 4,406 training and 1,886 test windows (attack rate 52.4%). The backdoor trigger is the CAN ID 0x7ff, verified absent from every capture.

Individual contribution. I wrote the entire `src/canids` package from scratch – the HCRL parser and time-contiguous windowing, the 1-D CNN (GroupNorm, so the same model trains under DP-SGD) and the RandomForest cross-check, the trigger-injection and poisoning code, the ASR measurement, the local-DP mechanisms and LDP Naive Bayes, the PCA/DCA compression, the output-modification defence and the membership-inference attack – plus the experiment driver and the unit tests. Opacus provides only the DP-SGD accountant.

4 Evaluation, Comparison and Critical Analysis

Results are averaged over seeds; $\delta = 10^{-5}$ for all DP.

4.1 Honest detector

Table 3 shows the baseline IDS detects CAN intrusions almost perfectly (CNN $F_1 = 0.996$). That near-perfect detector is precisely what the backdoor then defeats – the point of the study.

Table 3: Honest intrusion detectors.

Detector	Acc.	Bal. acc.	Prec.	Recall	F_1
RandomForest	0.946	0.948	1.000	0.897	0.946
1D-CNN	0.996	0.996	0.992	1.000	0.996

4.2 The backdoor (Module 1)

Figure 2 and Table 4 sweep the poison rate. The honest model ignores the trigger (ASR ≈ 0.000); once poisoned, the trigger flips genuine attacks to "Normal" at a high rate from a small poison fraction, while clean accuracy stays close to the honest baseline – an attack that is both effective and

hidden.

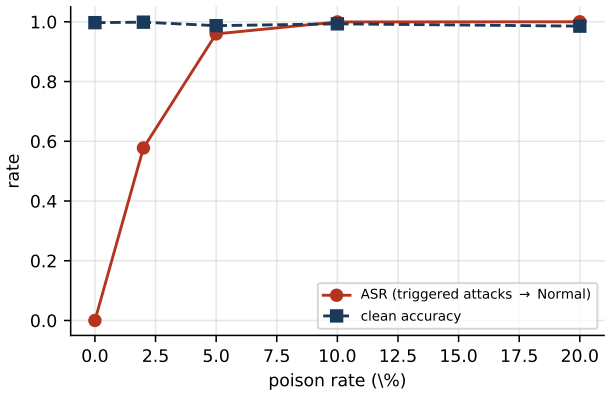


Figure 2: Clean accuracy and ASR vs poison rate.

Table 4: Backdoor poison sweep.

Poison rate	Clean acc.	ASR
0%	0.997	0.000
2%	0.999	0.578
5%	0.987	0.959
10%	0.993	0.999
20%	0.985	1.000

4.3 Differential privacy as a defence (Module 2)

Does private training blunt the backdoor? DP-SGD clips every window's gradient and adds noise, which is exactly the signal a few poisoned windows rely on. Figure 3 and Table 5 report, at a fixed poison rate, the detection F_1, the attack-success rate and the membership-inference AUC as ε varies. The finding is clear and nuanced: at moderate budgets ($\varepsilon \geq 4$) DP-SGD *eliminates* the backdoor (ASR falls to zero) while the detector stays strong ($F_1 \approx 0.98$) and membership inference stays near chance – a single mechanism that defeats the integrity attack and the privacy attack at once. Only at the tightest budget ($\varepsilon = 1$) does the detector itself degrade; its raised response to triggered inputs there reflects a loss of detection accuracy, not a surviving backdoor. This is the privacy–utility trade-off made concrete.

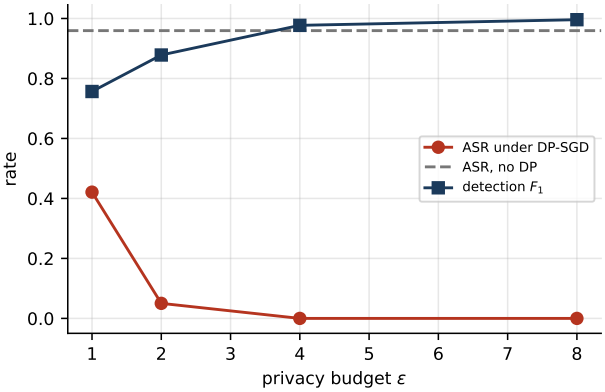


Figure 3: F_1 and ASR vs privacy budget ε.

Table 5: DP-SGD: utility, ASR, membership-inference AUC.

ε	F_1	ASR	MIA AUC
1.0	0.757	0.421	0.531
2.0	0.878	0.050	0.543
4.0	0.977	0.000	0.552
8.0	0.996	0.000	0.574
∞	0.983	0.959	0.643

DP-SGD also protects the training windows from membership inference, and a differentially private k-means over the windows shows the DP tax on unsupervised structure (Table 6).

Table 6: DP k-means agreement with the non-private clustering.

ε	ARI vs non-private
1.0	0.491
2.0	0.463
4.0	0.466
8.0	0.502

4.4 Local DP on the telemetry (Module 3)

If the frames are randomised at the source (before an untrusted curator sees them), the IDS still trains but loses some accuracy. Figure 4 sweeps the local budget for the Laplace, Duchi and Piecewise mechanisms, and Table 7 gives an LDP Naive Bayes detector.

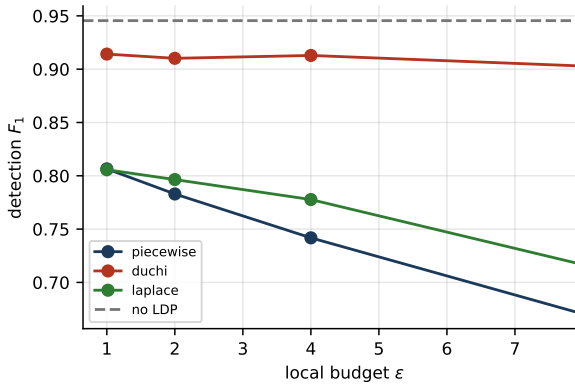


Figure 4: Detection F_1 under local-DP perturbat

Table 7: LDP Naive Bayes detector.

ε_{local}	LDP-NB F_1
1.0	0.853
2.0	0.807
4.0	0.822
8.0	0.844

4.5 Output modification and compressive privacy (Modules 4, 5)

Releasing only the hard label or a rounded confidence weakens the membership attack (Table 8). Compressing the 1100-dimensional windows with PCA or DCA keeps detection utility while discarding most reconstructable raw telemetry (Table 9, Fig. 5). Classical table k-anonymity transfers only partially to a CAN stream, so it is treated analytically (`can_id_kanonymity`: 46 distinct

Table 8: Output modification vs membership inference.

Released output	MIA AUC	MIA adv.
full-confidence	0.615	0.148
rounded-1dp	0.525	0.051
label-only	0.502	0.004

IDs).

Table 9: Compressive projections.

Method	p	detection F_1	recon. err.
PCA	20	0.825	0.917
DCA	20	0.921	0.971

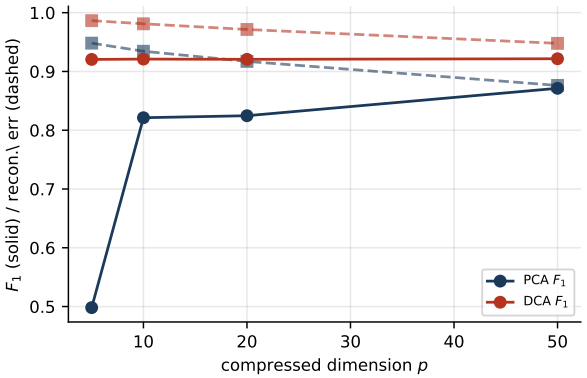


Figure 5: Detection F_1 and reconstruction error vs dimension.

4.6 Coverage synthesis

Table 10 consolidates which mechanism addresses which threat on the IDS.

Table 10: Threat–defence coverage (✓ stops, ∼ partial).

Threat	Mechanism	Stops?	Note
Backdoor poisoning (integrity)	DP-SGD training	∼	Clipping+noise limit poison influence; measured by ASR-vs-ε
Membership inference	DP-SGD training	✓	Bounds per-window influence
Membership inference	Output modification	∼	Removes the confidence signal
Telemetry disclosure at collection	Local DP perturbation	✓	No trusted curator
Raw-window reconstruction	Compressive PCA/DCA	✓	Low-dim code discards raw signal
Frame re-identification	CAN-ID k-anonymity (analytical)	∼	Stream, not a classic table

4.7 Limitations and validity

The mirror is a repackaged/relabelled copy, so the task is framed as binary normal-vs-attack; attack captures are wholly attacked, so the class balance is a sampling choice; and the honest detector is near-perfect because these CAN attacks are statistically blunt – which is the point (a near-perfect guard is silently defeated), not a flaw. None of these change the direction of the results.

5 Conclusions and Final Documentation

A trigger-based backdoor turns a near-perfect CAN intrusion detector into one that passes real attacks on command, while staying invisible on clean traffic. Differentially private training is the one mechanism that both protects the training windows and attacks the backdoor's own mechanism; local DP and compressive privacy protect the telemetry at a measured utility

cost. All results are reproducible from `IA2/scripts/run_all.py` (full codebase at `https://github.com/aish475/MLCS_IA`); a companion IEEE-format manuscript accompanies this report.

References

Tianyu Gu, Brendan Dolan-Gavitt, and Siddharth Garg. BadNets: Identifying vulnerabilities in the machine learning model supply chain. In *NIPS Machine Learning and Computer Security Workshop*, 2017.

Karl Koscher et al. Experimental security analysis of a modern automobile. *IEEE S&P*, 2010.